LESSON PLAN · GENERATIVE AI (GR 6–8) BEGINNER: GEN-AI CREATORS

Lesson 6: Teach a Machine

Module 2 | 60 minutes | Machine learning from data | Grades 6–11 · ages 11–16

Objective: Train an image model in Google Teachable Machine by collecting examples, then test it, find where it fails and explain why in a data diary.

Success criteria

- I can explain what machine learning is.
- I can collect data, train and test a model in Teachable Machine.
- I can find where my model fails and explain why.
- I can improve a model with better data.

Key vocabulary

- **machine learning**: When a computer learns patterns from examples.
- **class**: One group the model learns to recognise, like “Paper”.
- **data**: The examples you give the model to learn from.
- **train**: Let the computer study the examples and learn the pattern.
- **test**: Check the model with new examples it has never seen.
- **accuracy**: How often the model is right, e.g. 8 out of 10.

You will need

- Projector or screen share with Kitabash and the AI tools for this lesson
- Devices: 1 per learner (laptop, tablet or phone)
- Learner accounts only where allowed: see the course safety notes
- Printed worksheet (optional)
- Project files: l06-teach-a-machine-project-kit.md, l06-teacher-demo-script.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game**	Human classifier: learn “zibbles” from ten examples, then vote on tricky new items.
5 min	**Watch + thumbs check**	Play the video. Thumbs check: “If it works on my training photos, it is a good model.” (Thumbs down.)
8 min	**Live demo**	Teacher builds Paper / Plastic / Nothing on one desk, then breaks it with new tests and fixes it with varied data.
5 min	**Prompt Lab**	Learners write their data diary plan and tick every checklist item.

17 min	**Build (pairs)**	Pairs build a recycling sorter or rock-paper-scissors detector in Teachable Machine: collect, train, run 10 tests, record in the data diary. The ML loop order game while models train.
6 min	**Quick competition**	“Break my model” challenge: pairs swap screens and try to find a failure in another pair’s model in 2 minutes. Then the Helps or confuses sort.
6 min	**Show-and-tell**	Three pairs share their accuracy before and after improving, and the failure they found.
4 min	**Probe**	Why did it fail? and New situation: the crushed bottle.
4 min	**Exit ticket**	Exit ticket in the chat.

Unplugged warm-up

Human classifier (5 min): show the class five pictures of “zibbles” (all small, round, red items) and five “not zibbles”. Then show new items, including a big red round thing and a small round blue thing. Learners vote. Discuss: what rule did you learn from only ten examples? Would more varied examples help?

Teaching the idea

Machine learning lets a computer learn patterns from labelled examples instead of hand-written rules. In Teachable Machine (no account, runs in the browser) learners create classes, add examples by webcam or upload, train, and preview. The key lessons are about data: more varied, correctly labelled examples generalise better; a model learns whatever patterns are in the data — including background and lighting — and fails on what it has never seen. This is the same idea as bias in Lesson 5. Check moment: structured testing with new objects and a data diary. Privacy: film objects or hands only, never faces of other people; nothing needs to be uploaded or shared.

Analogy: Teaching a toddler what a dog is: if you only ever show them white dogs, they may say a black dog is a cat. Variety in examples makes better learners — human or machine.

Live demo: Share screen at teachablemachine.withgoogle.com. 1) Get Started → Image Project → standard image model. 2) Make classes “Paper”, “Plastic”, “Nothing”. Record about 30 webcam examples each, deliberately all on the same desk in the same light. 3) Train and test with the same objects — it looks perfect. 4) Now test a new object, a different background or dim light; find a mistake together and ask “Why?” 5) Add more varied examples, retrain and show the improvement. If the model happens to work well, say so honestly, then keep trying harder tests until you find a failure — every model has one. Show the data diary on screen as you go.

Questions to ask

Ask	Listen for
What is a class in Teachable Machine?	One group the model learns, like Paper or Plastic.

Why test with new objects?	To see if it learned the real pattern, not just your training photos.
Why might your model fail in dim light?	It never saw dim-light examples; it may use brightness as a clue.
How do you fix a model that fails?	Add more varied, correctly labelled data, train and test again.
How is this like bias in AI images?	Both learn from data; missing examples cause unfair or wrong results.

Differentiation

Support
Use just two classes (e.g. Rock and Paper) and give a printed data diary with the test rows ready to fill in.

Stretch
Add a fourth class and measure accuracy out of 20 tests. Which class is confused most often, and why?

Common misconceptions

Misconception	How to fix it
The computer understands what paper is	It finds patterns in pixels — including background and light.
If it works on my training photos, it works	Only testing with new examples tells you how good it is.
More photos of the same thing is always better	Variety matters more than repeats: different angles, backgrounds and light.
The model fails because the computer is stupid	It fails because something was missing from the data. That is fixable.

Assessment

Use the Prompt Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Your model fails on a green bottle. In one sentence: why, and how would you fix it?

Homework
Teach someone at home how your model works. Then think of one real-life AI (like photo search or face unlock) and write what data it might need to be fair to everyone.

Answer key — Kitabash activities

1. Prompt Lab: my data diary plan (perform · ai lab)

Checklist: Name your classes · How many examples · Make the data varied · Test with new things · A failure you expect

Strong example:

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My classes are Paper, Plastic and Nothing. I will collect 40 photos for each class with different objects, angles, backgrounds and lighting. I will test it with 10 new objects it has never seen, in daylight and in dim light. I expect it might confuse shiny paper with plastic, so I will add more shiny paper photos.
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2. The ML loop (perform · ordering)

Order: Choose your classes → Collect varied examples for each class → Train the model → Test it with new examples → Find the mistakes and add better data → Train and test again

3. Helps the model, or confuses it? (perform · categorize)

Helps the model: Photos from many different angles, A “Nothing” class with an empty background, Photos in bright light and dim light
Confuses the model: Every “Paper” photo taken on the same white desk, Putting a few plastic photos in the Paper class by mistake, Only 3 photos per class

4. Why did it fail? (perform · multiple choice)

Answer: It was trained only on Zoya’s hand, so it has not learned other hands

5. New situation: the crushed bottle (probe · multiple choice)

Answer: Add many photos of crushed, dirty and different plastic items to the Plastic class, then train and test again

Answer key — worksheet

1. **Key words:** machine learning = When a computer learns patterns from examples. · class = One group the model learns to recognise, like “Paper”. · data = The examples you give the model to learn from. · train = Let the computer study the examples and learn the pattern.
2. **Prompt Lab: my data diary plan:** Include: Name your classes; How many examples; Make the data varied; Test with new things; A failure you expect. Example: My classes are Paper, Plastic and Nothing. I will collect 40 photos for each class with different objects, angles, backgrounds and lighting. I will test it with 10 new objects it has never seen, in daylight and in dim light. I expect it might confuse shiny paper with plastic, so I will add more shiny paper photos.

3. **The ML loop:** 1 Choose your classes · 2 Collect varied examples for each class · 3 Train the model · 4 Test it with new examples · 5 Find the mistakes and add better data · 6 Train and test again
4. **Helps the model, or confuses it?:** Helps the model: Photos from many different angles, A “Nothing” class with an empty background, Photos in bright light and dim light · Confuses the model: Every “Paper” photo taken on the same white desk, Putting a few plastic photos in the Paper class by mistake, Only 3 photos per class
5. **Why did it fail?:** B. It was trained only on Zoya’s hand, so it has not learned other hands — A model only knows what it was shown. Add examples from other hands (with permission, and no faces) to make it fair for everyone.

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